CSE 615

Machine Learning

Quiz Bank — MCQ + Short Questions (with Answers)

Batch 1: MCQs + Short Questions

Organized by Part 1–6, following the same sequence as the course notes.

Part 1: Foundations and Data Preparation — MCQ

Q1. The main goal of a machine learning model is to:

A. Memorize training data exactly
B. Generalize well to new, unseen data
C. Run as slowly as possible
D. Avoid using any data
Answer: B. Generalize well to new, unseen data

Q2. Learning can be framed as approximating an unknown function f using a hypothesis h chosen from a:

A. Random number generator
B. Hypothesis space
C. Single fixed rule
D. Database index
Answer: B. Hypothesis space

Q3. Which of these is NOT typically part of the data preparation stage?

A. Data cleansing
B. Handling missing values
C. Deploying the model to production
D. Data exploration
Answer: C. Deploying the model to production

Q4. Imputing missing values with the median is an example of:

A. Data collection
B. Data cleansing
C. Model deployment
D. Feature scaling only
Answer: B. Data cleansing

Q5. Concept representation in ML refers to:

A. The dataset's file format
B. The internal form learned knowledge takes, e.g. a tree or weight vector
C. The programming language used
D. The company's business model
Answer: B. The internal form learned knowledge takes, e.g. a tree or weight vector

Q6. Why does real-world ML spend so much effort on data preparation?

A. Because algorithms are always perfect
B. Because model quality heavily depends on clean, well-structured input data
C. Because it is legally required
D. Because it makes training slower on purpose
Answer: B. Because model quality heavily depends on clean, well-structured input data

Part 1 — Short Questions

S1. What does it mean for a model to 'generalize'?

Ans: It means the model performs well on new, unseen data, not just the specific examples it was trained on.

S2. List four typical steps in data preparation.

Ans: Any four of: collection, discovery, exploration, cleansing, validation, structuring.

S3. Why is ML often framed as function approximation?

Ans: Because we assume there's a true (unknown) function mapping inputs to outputs, and learning tries to find a hypothesis that approximates this function using the training examples.

Part 2: Tree-Based and Rule-Based Learning — MCQ

Q7. A decision tree splits data at each internal node based on:

A. Random guesses
B. A test on a feature value
C. The final prediction only
D. The tree's depth
Answer: B. A test on a feature value

Q8. Which measure is commonly used to choose the best attribute to split on?

A. File size
B. Information gain (entropy reduction)
C. Alphabetical order
D. Number of leaves
Answer: B. Information gain (entropy reduction)

Q9. Random Forest reduces overfitting mainly by:

A. Using a single very deep tree
B. Averaging/voting over many diverse trees trained on random subsets
C. Removing all randomness
D. Ignoring the training data
Answer: B. Averaging/voting over many diverse trees trained on random subsets

Q10. Boosting methods like AdaBoost build trees:

A. All independently and simultaneously
B. Sequentially, each focusing on correcting previous errors
C. Only once, never iterating
D. Without using any weights
Answer: B. Sequentially, each focusing on correcting previous errors

Q11. A sign that a decision tree is overfitting is:

A. High training accuracy but low validation/test accuracy
B. Low accuracy everywhere
C. Equal accuracy on training and test data always
D. The tree has no leaves
Answer: A. High training accuracy but low validation/test accuracy

Q12. Pruning a decision tree means:

A. Adding more branches
B. Removing branches that don't improve validation performance
C. Deleting the entire tree
D. Increasing tree depth
Answer: B. Removing branches that don't improve validation performance

Q13. A propositional rule differs from a first-order rule because it:

A. Uses variables and relations between objects
B. Uses simple fixed attribute-value conditions without variables
C. Cannot be used in ML at all
D. Always outputs probabilities
Answer: B. Uses simple fixed attribute-value conditions without variables

Q14. Sequential covering is a strategy that:

A. Learns all rules simultaneously
B. Learns one good rule at a time, removing covered examples before learning the next
C. Never removes any examples
D. Only works for regression
Answer: B. Learns one good rule at a time, removing covered examples before learning the next

Q15. Every decision tree can be converted into:

A. A neural network only
B. An equivalent set of if-then rules, one per root-to-leaf path
C. A clustering algorithm
D. A single rule covering everything
Answer: B. An equivalent set of if-then rules, one per root-to-leaf path

Q16. Inverse resolution, from Inductive Logic Programming, works by:

A. Working forward from rules to conclusions only
B. Working backward from resolution-based inference to generate a hypothesis explaining the examples
C. Ignoring background knowledge entirely
D. Only applying to numeric data
Answer: B. Working backward from resolution-based inference to generate a hypothesis explaining the examples

Part 2 — Short Questions

S4. What is the difference between a decision tree and a regression tree?

Ans: A decision (classification) tree predicts a discrete class label at each leaf; a regression tree predicts a continuous numeric value at each leaf.

S5. Explain the core idea behind ensemble methods like Random Forest.

Ans: Combine many individual models (e.g., trees trained on random subsets of data/features) and aggregate their predictions (voting/averaging), which typically reduces variance and improves accuracy over any single model.

S6. How does gradient boosting differ from random forest in how trees are built?

Ans: Random forest builds trees independently and in parallel on random subsets; gradient boosting builds trees sequentially, each new tree trained to correct the residual errors of the combined previous trees.

S7. Why is overfitting a concern for decision trees specifically?

Ans: Because trees can keep splitting until every training example (including noise) is perfectly classified, creating overly specific rules that don't generalize to new data.

Part 3: Regression Analysis and Bayesian Learning — MCQ

Q17. Simple linear regression models the relationship between:

A. Multiple inputs and multiple outputs
B. One input and one output, as a straight line
C. Only categorical variables
D. Clusters of data
Answer: B. One input and one output, as a straight line

Q18. Linear regression weights are typically chosen to minimize:

A. The number of features
B. The sum of squared prediction errors (least squares)
C. The training time
D. The number of training examples
Answer: B. The sum of squared prediction errors (least squares)

Q19. Logistic regression is primarily used for:

A. Regression of continuous values
B. Binary/multi-class classification, via a probability output
C. Clustering
D. Dimensionality reduction
Answer: B. Binary/multi-class classification, via a probability output

Q20. The logistic (sigmoid) function squashes its input into the range:

A. $(-\infty, \infty)$
B. (0, 1)
C. (-1, 1)
D. [0, 100]
Answer: B. (0, 1)

Q21. Bayes' Theorem is written as P(h|D) = ?

A. P(D|h) / P(h)
B. P(D|h)P(h) / P(D)
C. P(h) + P(D)
D. P(h) × P(D) + 1
Answer: B. P(D|h)P(h) / P(D)

Q22. Naive Bayes makes the simplifying assumption that:

A. All features are perfectly correlated
B. Features are conditionally independent given the class
C. There is only one feature
D. The prior probability is always 0.5
Answer: B. Features are conditionally independent given the class

Q23. Bayesian Optimization is commonly used for:

A. Cleaning missing data
B. Efficiently searching for good hyperparameters using a probabilistic model
C. Drawing decision trees
D. Computing entropy
Answer: B. Efficiently searching for good hyperparameters using a probabilistic model

Q24. A Bayesian Belief Network represents:

A. A single decision tree
B. A joint probability distribution over variables as a directed graph of dependencies
C. A clustering algorithm
D. A neural network layer
Answer: B. A joint probability distribution over variables as a directed graph of dependencies

Q25. Multiple linear regression differs from simple linear regression by:

A. Using zero input features
B. Using several input features combined linearly to predict the output
C. Only working for classification
D. Removing the need for training data
Answer: B. Using several input features combined linearly to predict the output

Part 3 — Short Questions

S8. Write the general form of a multiple linear regression model.

Ans: y = w0 + w1x1 + w2x2 + ... + wnxn, where wi are learned weights and xi are input features.

S9. Why is logistic regression used for classification rather than regression, despite its name?

Ans: Because it models the probability of a class label using the sigmoid function applied to a linear combination of features, then applies a threshold to output a discrete class, rather than predicting an unbounded continuous value.

S10. State Bayes' Theorem and explain each term briefly.

Ans: P(h|D) = P(D|h)P(h)/P(D): P(h|D) is the posterior belief in hypothesis h after seeing data D; P(D|h) is the likelihood; P(h) is the prior; P(D) is the evidence/normalizing constant.

S11. Why is Naive Bayes called 'naive'?

Ans: Because it assumes all features are conditionally independent given the class label, an assumption that is often not strictly true in real data but still works well in practice.

Part 4: Neural Networks and SVMs — MCQ

Q26. A linear threshold unit outputs 1 when:

A. The weighted sum of inputs is below the threshold
B. The weighted sum of inputs exceeds the threshold
C. There are no inputs
D. The bias is zero
Answer: B. The weighted sum of inputs exceeds the threshold

Q27. The perceptron learning rule updates weights based on:

A. Random noise
B. The error on misclassified examples
C. The total number of layers
D. The dataset's file size
Answer: B. The error on misclassified examples

Q28. A single perceptron cannot learn which classic function?

A. AND
B. OR
C. XOR
D. NOT
Answer: C. XOR

Q29. Backpropagation computes gradients by:

A. Ignoring the chain rule
B. Propagating the error backward from output to hidden layers using the chain rule
C. Only updating the output layer
D. Randomly guessing weight updates
Answer: B. Propagating the error backward from output to hidden layers using the chain rule

Q30. Why can multilayer networks learn XOR while a single perceptron cannot?

A. Multilayer networks have no weights
B. Hidden layers with non-linear activations let the network represent non-linear decision boundaries
C. XOR is actually linearly separable
D. Multilayer networks are slower only
Answer: B. Hidden layers with non-linear activations let the network represent non-linear decision boundaries

Q31. Recurrent Neural Networks are especially suited for:

A. Independent, unordered data points
B. Sequential data, since they carry forward a hidden state over time
C. Only image classification
D. Data with no temporal structure
Answer: B. Sequential data, since they carry forward a hidden state over time

Q32. An SVM's 'support vectors' are:

A. All the training points
B. The training points closest to the decision boundary that define the margin
C. Random noise points
D. Only points from one class
Answer: B. The training points closest to the decision boundary that define the margin

Q33. The SVM optimization problem for finding the maximum margin separator is formulated as:

A. A linear search
B. A quadratic programming problem
C. A sorting algorithm
D. A clustering step
Answer: B. A quadratic programming problem

Q34. The kernel trick in SVMs allows:

A. Only linear classification
B. Implicit mapping to a higher-dimensional space for non-linear separation, without explicitly computing that mapping
C. Removing all support vectors
D. Training without any data
Answer: B. Implicit mapping to a higher-dimensional space for non-linear separation, without explicitly computing that mapping

Q35. Dropout is a technique used to:

A. Increase overfitting
B. Reduce overfitting in neural networks by randomly disabling units during training
C. Speed up data collection
D. Replace backpropagation entirely
Answer: B. Reduce overfitting in neural networks by randomly disabling units during training

Part 4 — Short Questions

S12. Why is XOR a classic example used to motivate multilayer neural networks?

Ans: Because XOR is not linearly separable, so a single perceptron cannot learn it, but a network with a hidden layer can, by learning intermediate non-linear features that the output layer combines.

S13. Explain what the 'margin' means in the context of SVMs.

Ans: The margin is the distance between the separating hyperplane and the closest training points (support vectors) of either class; SVM chooses the hyperplane that maximizes this margin.

S14. What problem does the kernel trick solve for SVMs?

Ans: It lets SVMs classify data that isn't linearly separable in the original feature space, by implicitly operating in a higher-dimensional space where a linear separator exists, without the computational cost of explicitly transforming the data.

S15. Briefly describe the backpropagation algorithm.

Ans: It computes the gradient of the network's error with respect to every weight by propagating error signals backward from the output layer through each hidden layer (using the chain rule), then updates weights via gradient descent.

Part 5: Instance-Based Learning and Clustering — MCQ

Q36. K-Nearest Neighbor classifies a new point based on:

A. A pre-trained explicit model only
B. The majority class among its K closest training examples
C. Random assignment
D. The furthest training example
Answer: B. The majority class among its K closest training examples

Q37. Instance-based learning is described as 'lazy' because it:

A. Trains for a very long time upfront
B. Defers computation until prediction time, using stored instances directly
C. Never makes predictions
D. Requires no data at all
Answer: B. Defers computation until prediction time, using stored instances directly

Q38. A Self-Organizing Map (SOM) is used mainly to:

A. Perform supervised classification only
B. Project high-dimensional data onto a low-dimensional grid, preserving topology
C. Replace decision trees
D. Compute Bayes' theorem
Answer: B. Project high-dimensional data onto a low-dimensional grid, preserving topology

Q39. Case-Based Reasoning solves new problems by:

A. Ignoring past experience entirely
B. Retrieving and adapting solutions to similar past cases
C. Random guessing
D. Always using the same fixed rule
Answer: B. Retrieving and adapting solutions to similar past cases

Q40. k-means clustering repeatedly performs which two steps?

A. Splitting and merging clusters randomly
B. Assigning points to nearest center, then recomputing centers as means
C. Sorting and filtering
D. Only recomputing centers, never reassigning points
Answer: B. Assigning points to nearest center, then recomputing centers as means

Q41. DBSCAN's main advantage over k-means is that it:

A. Requires specifying the number of clusters in advance
B. Does not require specifying the number of clusters, and can identify noise/outliers
C. Only works on 1D data
D. Always produces spherical clusters
Answer: B. Does not require specifying the number of clusters, and can identify noise/outliers

Q42. Agglomerative hierarchical clustering starts with:

A. One giant cluster containing everything
B. Every point as its own cluster, then merges closest pairs
C. A fixed number of random clusters
D. No data at all
Answer: B. Every point as its own cluster, then merges closest pairs

Q43. t-SNE is primarily used for:

A. Classification
B. Visualizing high-dimensional data by preserving local neighborhood structure in 2D/3D
C. Regression
D. Rule induction
Answer: B. Visualizing high-dimensional data by preserving local neighborhood structure in 2D/3D

Q44. PCA reduces dimensionality by projecting data onto:

A. Randomly chosen axes
B. Directions of greatest variance (principal components)
C. The mean of the dataset only
D. A single fixed axis always
Answer: B. Directions of greatest variance (principal components)

Q45. The Expectation-Maximization (EM) algorithm is commonly used to fit:

A. Decision trees
B. Gaussian Mixture Models for probabilistic clustering
C. Support vector machines
D. Linear regression
Answer: B. Gaussian Mixture Models for probabilistic clustering

Part 5 — Short Questions

S16. What does it mean that K-NN is a 'lazy' learner?

Ans: It doesn't build an explicit model during training; instead it stores the training data and does all the real computation (finding nearest neighbors) at prediction time.

S17. Explain the two repeated steps of the k-means algorithm.

Ans: 1) Assign each point to its nearest of the k current cluster centers. 2) Recompute each center as the mean of the points currently assigned to it. Repeat until assignments stabilize.

S18. Why doesn't DBSCAN require specifying the number of clusters in advance?

Ans: Because it forms clusters based on regions of sufficient point density, discovering the number of clusters naturally from the data's density structure, rather than needing a predefined k.

S19. What is the difference between MDS/t-SNE and PCA?

Ans: PCA is a linear technique projecting onto directions of maximum variance; MDS and t-SNE are typically non-linear techniques focused on preserving pairwise distances (MDS) or local neighborhood structure (t-SNE) for visualization.

Part 6: Experimental Evaluation — MCQ

Q46. Accuracy alone can be misleading when:

A. Classes are perfectly balanced
B. Classes are highly imbalanced, e.g. a rare disease
C. The dataset is very large
D. There is no noise in the data
Answer: B. Classes are highly imbalanced, e.g. a rare disease

Q47. The F1 score is the:

A. Sum of precision and recall
B. Harmonic mean of precision and recall
C. Difference between accuracy and recall
D. Product of accuracy and precision
Answer: B. Harmonic mean of precision and recall

Q48. In k-fold cross-validation, each data point is used for testing:

A. Never
B. Exactly once, across the k rotations
C. k times
D. Only if it's an outlier
Answer: B. Exactly once, across the k rotations

Q49. The bootstrap method creates new training sets by:

A. Removing all duplicate data
B. Resampling the dataset with replacement
C. Always using the full original dataset unchanged
D. Deleting half the data randomly
Answer: B. Resampling the dataset with replacement

Q50. A learning curve plots performance against:

A. The number of features only
B. The amount of training data used
C. The model's file size
D. The programming language used
Answer: B. The amount of training data used

Q51. If a learning curve is still rising steeply, it suggests:

A. More data would likely NOT help
B. More training data would likely improve performance further
C. The model is already perfect
D. The model has too many features
Answer: B. More training data would likely improve performance further

Q52. A noise curve is used to study:

A. How training time changes with data size
B. How performance degrades as artificial noise is added to data
C. How many clusters exist
D. The model's memory usage
Answer: B. How performance degrades as artificial noise is added to data

Q53. Statistical hypothesis testing (e.g., paired t-test) is used to check whether:

A. A model has zero error always
B. An observed performance difference between two algorithms is statistically significant
C. A dataset has missing values
D. A feature is categorical
Answer: B. An observed performance difference between two algorithms is statistically significant

Q54. The holdout method's main limitation compared to cross-validation is:

A. It uses too much computation
B. Its performance estimate can vary a lot depending on the single random split chosen
C. It cannot be used at all
D. It requires no test data
Answer: B. Its performance estimate can vary a lot depending on the single random split chosen

Part 6 — Short Questions

S20. Why might a model with high accuracy still be a poor classifier?

Ans: If classes are imbalanced, a model can achieve high accuracy by mostly predicting the majority class while performing very poorly (low recall/precision) on the minority class that actually matters.

S21. Explain k-fold cross-validation and why it's preferred over a single holdout split.

Ans: The data is split into k folds; the model trains on k-1 folds and tests on the remaining fold, rotating through all k folds so every point is tested once; averaging results over all folds gives a more stable, reliable performance estimate than one random holdout split.

S22. What does a learning curve help you decide?

Ans: Whether collecting more training data is likely to improve model performance (curve still rising) or whether performance has plateaued (curve flattened), guiding whether more data collection is worthwhile.

S23. Why use statistical hypothesis testing when comparing two ML algorithms?

Ans: Because a small observed difference in test performance could simply be due to random variation in the particular test data, not a real difference in the algorithms; hypothesis testing checks whether the difference is likely genuine.

End of Batch 1 (CSE 615)

This batch contains 54 MCQs and 23 short questions, covering Parts 1-6 in sequence. Ask for more batches to keep building toward the full 500-question target.